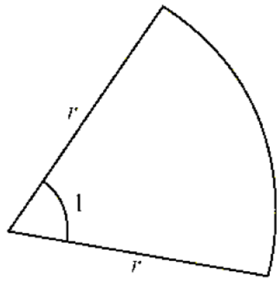
3-(MAT/11_HL_Summer_2019_Q3)-*Radian*

[Maximum mark: 4]

A sector of a circle with radius r cm, where $r > 0$, is shown on the following diagram. The sector has an angle of 1 radian at the centre.



Let the area of the sector be $A\text{ cm}^2$ and the perimeter be $P\text{ cm}$. Given that $A = P$, find the value of r.

1-(MAT/11_HL_Summer_2012_Q5)-*Trigonometry*

Let $f(x) = \dfrac{\sin 3x}{\sin x} - \dfrac{\cos 3x}{\cos x}$.

(a) For what values of x does $f(x)$ not exist? *[2 marks]*

(b) Simplify the expression $\dfrac{\sin 3x}{\sin x} - \dfrac{\cos 3x}{\cos x}$. *[5 marks]*

2-(MAT/12_HL_Summer_2012_Q9)**-***Trigonometry*

[Maximum mark: 6]

Show that $\dfrac{\cos A+\sin A}{\cos A-\sin A}=\sec 2A+\tan 2A$.

3–(MAT/11_HL_Summer_2012_Q10)**–***Trigonometry*

In the triangle ABC, $A\hat{B}C = 90^\circ$, $AC = \sqrt{2}$ and $AB = BC + 1$.

(a) Show that $\cos\hat{A} - \sin\hat{A} = \frac{1}{\sqrt{2}}$. *[3 marks]*

(b) By squaring both sides of the equation in part (a), solve the equation to find the angles in the triangle. *[8 marks]*

(c) Apply Pythagoras' theorem in the triangle ABC to find BC, and hence show that $\sin\hat{A} = \frac{\sqrt{6}-\sqrt{2}}{4}$. *[6 marks]*

(d) Hence, or otherwise, calculate the length of the perpendicular from B to [AC]. *[4 marks]*

4-(MAT/10_HL_Winter_2012_Q1)-*Trigonometry*

[Maximum mark: 4]

Given that $\frac{\pi}{2} < \alpha < \pi$ and $\cos\alpha = -\frac{3}{4}$, find the value of $\sin 2\alpha$.

5-(MAT/10_HL_Winter_2012_Q7)-*Trigonometry*

In the triangle PQR, $PQ = 6$, $PR = k$ and $P\hat{Q}R = 30^\circ$.

(a) For the case $k = 4$, find the two possible values of QR. *[4 marks]*

(b) Determine the values of k for which the conditions above define a unique triangle. *[3 marks]*

6-(MAT/12_HL_Summer_2013_Q10)**-***Trigonometry*

(a) Given that $\arctan\left(\frac{1}{5}\right)+\arctan\left(\frac{1}{8}\right)=\arctan\left(\frac{1}{p}\right)$, where $p \in \mathbb{Z}^+$, find p. *[3 marks]*

(b) Hence find the value of $\arctan\left(\frac{1}{2}\right)+\arctan\left(\frac{1}{5}\right)+\arctan\left(\frac{1}{8}\right)$. *[3 marks]*

7–(MAT/11_HL_Summer_2013_Q11)**–***Trigonometry*

(a) (i) Express $\cos\left(\frac{\pi}{6}+x\right)$ in the form $a\cos x-b\sin x$ where $a,\ b\in\mathbb{R}$.

(ii) Hence solve $\sqrt{3}\cos x-\sin x=1$ for $0\le x\le 2\pi$. *[7 marks]*

(b) Let $p(x)=2x^3-x^2-2x+1$.

(i) Show that $x=1$ is a zero of p.

(ii) Hence find all the solutions of $2x^3-x^2-2x+1=0$.

(iii) Express $\sin 2\theta\cos\theta+\sin^2\theta$ in terms of $\sin\theta$.

(iv) Hence solve $\sin 2\theta\cos\theta+\sin^2\theta=1$ for $0\le\theta\le 2\pi$. *[12 marks]*

8-(MAT/10_HL_Winter_2013_Q8)**-***Trigonometry*

(a) Prove the trigonometric identity $\sin(x+y)\sin(x-y) = \sin^2 x - \sin^2 y$. *[4]*

(b) Given $f(x) = \sin\left(x+\frac{\pi}{6}\right)\sin\left(x-\frac{\pi}{6}\right)$, $x \in [0, \pi]$, find the range of f. *[2]*

(c) Given $g(x) = \csc\left(x+\frac{\pi}{6}\right)\csc\left(x-\frac{\pi}{6}\right)$, $x \in [0, \pi]$, $x \neq \frac{\pi}{6}$, $x \neq \frac{5\pi}{6}$, find the range of g. *[2]*

9–(MAT/10_HL_Winter_2013_Q12)–*ComplexNumbers,Trigonometry,Integration*

Consider the complex number $z = \cos\theta + \mathrm{i}\sin\theta$.

(a) Use De Moivre's theorem to show that $z^n + z^{-n} = 2\cos n\theta$, $n \in \mathbb{Z}^+$. *[2]*

(b) Expand $\left(z + z^{-1}\right)^4$. *[1]*

(c) Hence show that $\cos^4\theta = p\cos 4\theta + q\cos 2\theta + r$, where p, q and r are constants to be determined. *[4]*

(d) Show that $\cos^6\theta = \frac{1}{32}\cos 6\theta + \frac{3}{16}\cos 4\theta + \frac{15}{32}\cos 2\theta + \frac{5}{16}$. *[3]*

(e) Hence find the value of $\int_0^{\frac{\pi}{2}} \cos^6\theta \,\mathrm{d}\theta$. *[3]*

The region S is bounded by the curve $y = \sin x\cos^2 x$ and the x-axis between $x = 0$ and $x = \frac{\pi}{2}$.

(f) S is rotated through 2π radians about the x-axis. Find the value of the volume generated. *[4]*

(g) (i) Write down an expression for the constant term in the expansion of $\left(z + z^{-1}\right)^{2k}$, $k \in \mathbb{Z}^+$.

(ii) Hence determine an expression for $\int_0^{\frac{\pi}{2}} \cos^{2k}\theta \,\mathrm{d}\theta$ in terms of k. *[3]*

10–(MAT/11_HL_Summer_2014_Q5)–*Trigonometry*

(a) Use the identity $\cos 2\theta = 2\cos^2\theta - 1$ to prove that $\cos\frac{1}{2}x = \sqrt{\frac{1+\cos x}{2}}$, $0 \le x \le \pi$. *[2]*

(b) Find a similar expression for $\sin\frac{1}{2}x$, $0 \le x \le \pi$. *[2]*

(c) Hence find the value of $\int_0^{\frac{\pi}{2}}\left(\sqrt{1+\cos x}+\sqrt{1-\cos x}\right)\mathrm{d}x$. *[4]*

11–(MAT/12_HL_Summer_2014_Q5)–*Graphs,Trigonometry*

(a) Sketch the graph of $y = \left|\cos\left(\frac{x}{4}\right)\right|$ for $0 \le x \le 8\pi$. *[2]*

(b) Solve $\left|\cos\left(\frac{x}{4}\right)\right| = \frac{1}{2}$ for $0 \le x \le 8\pi$. *[3]*

12-(MAT/11_HL_Summer_2014_Q7)-*Trigonometry*

[Maximum mark: 5]

The triangle ABC is equilateral of side 3 cm. The point D lies on [BC] such that $\text{BD} = 1\,\text{cm}$.
Find $\cos \text{D}\hat{\text{A}}\text{C}$.

13–(MAT/12_HL_Summer_2014_Q9)–*Sequences-Series,Trigonometry*

The first three terms of a geometric sequence are $\sin x$, $\sin 2x$ and $4\sin x\cos^2 x$, $-\frac{\pi}{2} < x < \frac{\pi}{2}$.

(a) Find the common ratio r. *[1]*

(b) Find the set of values of x for which the geometric series $\sin x + \sin 2x + 4\sin x\cos^2 x + \ldots$ converges. *[3]*

Consider $x = \arccos\left(\frac{1}{4}\right)$, $x > 0$.

(c) Show that the sum to infinity of this series is $\frac{\sqrt{15}}{2}$. *[3]*

14-(MAT/11_HL_Summer_2014_Q10)-*Trigonometry*

[Maximum mark: 6]

Given that $\sin x + \cos x = \frac{2}{3}$, find $\cos 4x$.

15-(MAT/10_HL_Winter_2014_Q13)-*ComplexNumbers,Trigonometry,Integration*

(a) (i) Show that $(1+\mathrm{i}\tan\theta)^n+(1-\mathrm{i}\tan\theta)^n=\dfrac{2\cos n\theta}{\cos^n\theta}$, $\cos\theta\neq 0$.

(ii) Hence verify that $\mathrm{i}\tan\dfrac{3\pi}{8}$ is a root of the equation $(1+z)^4+(1-z)^4=0$, $z\in\mathbb{C}$.

(iii) State another root of the equation $(1+z)^4+(1-z)^4=0$, $z\in\mathbb{C}$. *[10]*

(b) (i) Use the double angle identity $\tan 2\theta=\dfrac{2\tan\theta}{1-\tan^2\theta}$ to show that $\tan\dfrac{\pi}{8}=\sqrt{2}-1$.

(ii) Show that $\cos 4x=8\cos^4 x-8\cos^2 x+1$.

(iii) Hence find the value of $\displaystyle\int_0^{\frac{\pi}{8}}\frac{2\cos 4x}{\cos^2 x}\,\mathrm{d}x$. *[13]*

16–(MAT/12_HL_Summer_2015_Q3)–*Trigonometry*

[Maximum mark: 6]

Find all solutions to the equation $\tan x + \tan 2x = 0$ where $0^\circ \leq x < 360^\circ$.

17–(MAT/12_HL_Summer_2015_Q6)–*Trigonometry*

In triangle ABC, $\mathrm{BC} = \sqrt{3}\,\mathrm{cm}$, $\mathrm{A\hat{B}C} = \theta$ and $\mathrm{B\hat{C}A} = \frac{\pi}{3}$.

(a) Show that length $\mathrm{AB} = \dfrac{3}{\sqrt{3}\cos\theta + \sin\theta}$. [4]

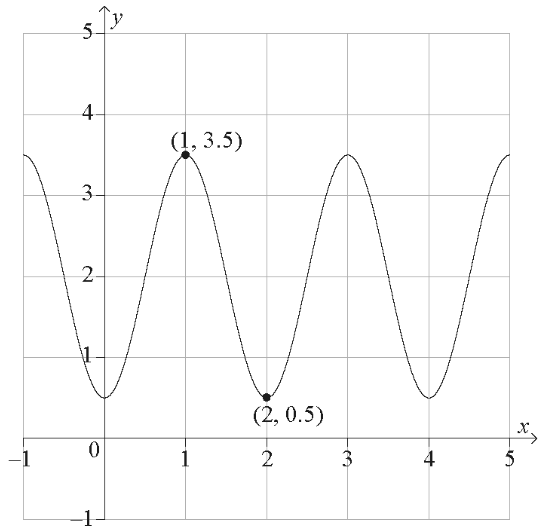
(b) Given that AB has a minimum value, determine the value of θ for which this occurs. [4]

18-(MAT/10_HL_Winter_2015_Q9)-*Trigonometry*

Solve the equation $\sin 2x - \cos 2x = 1 + \sin x - \cos x$ for $x \in [-\pi, \pi]$.

19–(MAT/11_HL_Summer_2016_Q3)–*Trigonometry*

The following diagram shows the curve $y = a\sin\left(b(x+c)\right)+d$, where a, b, c and d are all positive constants. The curve has a maximum point at $(1, 3.5)$ and a minimum point at $(2, 0.5)$.



(a) Write down the value of a and the value of d. [2]

(b) Find the value of b. [2]

(c) Find the smallest possible value of c, given $c > 0$. [2]

20–(MAT/12_HL_Summer_2016_Q3)–*Trigonometry,Integration*

(a) Show that $\cot\alpha = \tan\left(\frac{\pi}{2} - \alpha\right)$ for $0 < \alpha < \frac{\pi}{2}$. [1]

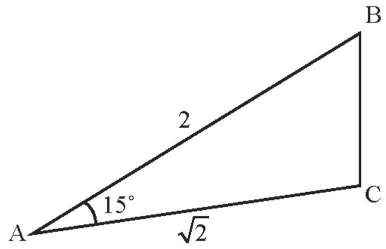
(b) Hence find $\displaystyle\int_{\tan\alpha}^{\cot\alpha} \frac{1}{1+x^2}\,\mathrm{d}x$, $0 < \alpha < \frac{\pi}{2}$. [4]

21–(MAT/11_HL_Summer_2016_Q5)–*Trigonometry*

(a) Expand and simplify $\left(1-\sqrt{3}\right)^2$. [1]

(b) By writing 15° as $60^\circ - 45^\circ$ find the value of $\cos(15^\circ)$. [3]

The following diagram shows the triangle ABC where $\mathrm{AB} = 2$, $\mathrm{AC} = \sqrt{2}$ and $\mathrm{B\hat{A}C} = 15^\circ$.



(c) Find BC in the form $a + \sqrt{b}$ where $a, b \in \mathbb{Z}$. [4]

22-(MAT/12_HL_Summer_2016_Q9)-*Trigonometry*

Consider the equation $\frac{\sqrt{3}-1}{\sin x}+\frac{\sqrt{3}+1}{\cos x}=4\sqrt{2},\ 0<x<\frac{\pi}{2}$. Given that $\sin\left(\frac{\pi}{12}\right)=\frac{\sqrt{6}-\sqrt{2}}{4}$ and $\cos\left(\frac{\pi}{12}\right)=\frac{\sqrt{6}+\sqrt{2}}{4}$

(a) verify that $x=\frac{\pi}{12}$ is a solution to the equation; [3]

(b) hence find the other solution to the equation for $0<x<\frac{\pi}{2}$. [5]

23-(MAT/10_HL_Winter_2016_Q13)-*Trigonometry*

(a) Find the value of $\sin\frac{\pi}{4}+\sin\frac{3\pi}{4}+\sin\frac{5\pi}{4}+\sin\frac{7\pi}{4}+\sin\frac{9\pi}{4}$. [2]

(b) Show that $\frac{1-\cos 2x}{2\sin x}\equiv \sin x,\ x\neq k\pi$ where $k\in\mathbb{Z}$. [2]

(c) Use the principle of mathematical induction to prove that
$\sin x+\sin 3x+\ldots+\ \sin(2n-1)x=\frac{1-\cos 2nx}{2\sin x},\ n\in\mathbb{Z}^{+},\ x\neq k\pi$ where $k\in\mathbb{Z}$. [9]

(d) Hence or otherwise solve the equation $\sin x+\sin 3x=\cos x$ in the interval $0<x<\pi$. [6]

24-(MAT/11_HL_Summer_2017_Q3)-*Trigonometry*

[Maximum mark: 5]

Solve the equation $\sec^2 x + 2\tan x = 0$, $0 \le x \le 2\pi$.

25–(MAT/12_HL_Summer_2017_Q11)–*Trigonometry,ComplexNumbers*

(a) Solve $2\sin(x+60^\circ)=\cos(x+30^\circ)$, $0^\circ \le x \le 180^\circ$. [5]

(b) Show that $\sin 105^\circ + \cos 105^\circ = \dfrac{1}{\sqrt{2}}$. [3]

(c) Let $z = 1-\cos 2\theta - \mathrm{i}\sin 2\theta$, $z \in \mathbb{C}$, $0 \le \theta \le \pi$.

(i) Find the modulus and argument of z in terms of θ. Express each answer in its simplest form.

(ii) Hence find the cube roots of z in modulus-argument form. [14]

26-(MAT/11_HL_Summer_2018_Q8)-*Trigonometry*

[Maximum mark: 5]

Let $a = \sin b$, $0 < b < \frac{\pi}{2}$.

Find, in terms of b, the solutions of $\sin 2x = -a$, $0 \le x \le \pi$.

27–(MAT/10_HL_Winter_2018_Q3)–*Trigonometry*

Consider the function $g(x) = 4\cos x + 1$, $a \le x \le \frac{\pi}{2}$ where $a < \frac{\pi}{2}$.

(a) For $a = -\frac{\pi}{2}$, sketch the graph of $y = g(x)$. Indicate clearly the maximum and minimum values of the function. [3]

(b) Write down the least value of a such that g has an inverse. [1]

(c) For the value of a found in part (b),

(i) write down the domain of g^{-1};

(ii) find an expression for $g^{-1}(x)$. [3]

28-(MAT/11_HL_Summer_2019_Q4)-*Trigonometry*

The lengths of two of the sides in a triangle are $4\,\text{cm}$ and $5\,\text{cm}$. Let θ be the angle between the two given sides. The triangle has an area of $\dfrac{5\sqrt{15}}{2}\,\text{cm}^2$.

(a) Show that $\sin\theta = \dfrac{\sqrt{15}}{4}$. [1]

(b) Find the two possible values for the length of the third side. [6]

29-(MAT/11_HL_Summer_2019_Q9)-*Trigonometry,Integration*

(a) Show that $(\sin x + \cos x)^2 = 1 + \sin 2x$. [2]

(b) Show that $\sec 2x + \tan 2x = \dfrac{\cos x + \sin x}{\cos x - \sin x}$. [4]

(c) Hence or otherwise find $\int_0^{\frac{\pi}{6}} (\sec 2x + \tan 2x)\,dx$ in the form $\ln\left(a + \sqrt{b}\right)$ where $a, b \in \mathbb{Z}$. [9]

30-(MAT/10_HL_Winter_2019_Q4)-*Trigonometry*

[Maximum mark: 7]

A and B are acute angles such that $\cos A = \frac{2}{3}$ and $\sin B = \frac{1}{3}$.

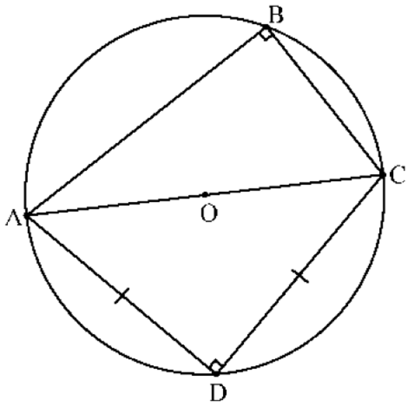
Show that $\cos(2A + B) = -\frac{2\sqrt{2}}{27} - \frac{4\sqrt{5}}{27}$.

31-(MAT/10_HL_Winter_2019_Q9)-*Trigonometry*

[Maximum mark: 14]

(a) Given that $\cos 75° = q$, show that $\cos 105° = -q$. [1]

In the following diagram, the points A, B, C and D are on the circumference of a circle with centre O and radius r. [AC] is a diameter of the circle. $\mathrm{BC} = r$, $\mathrm{AD} = \mathrm{CD}$ and $\mathrm{A\hat{B}C} = \mathrm{A\hat{D}C} = 90$.



(b) Show that $\mathrm{B\hat{A}D} = 75°$. [3]

(c) (i) By considering triangle ABD, show that $\mathrm{BD}^2 = 5r^2 - 2r^2 q\sqrt{6}$.

(ii) By considering triangle CBD, find another expression for BD^2 in terms of r and q. [7]

(d) Use your answers to part (c) to show that $\cos 75 = \dfrac{1}{\sqrt{6}+\sqrt{2}}$. [3]

32-(MAT/10_HL_Winter_2020_Q6)-*Quadratics,Trigonometry*

[Maximum mark: 4]

Consider the equation $ax^2 + bx + c = 0$, where $a \neq 0$. Given that the roots of this equation are $x = \sin\theta$ and $x = \cos\theta$, show that $b^2 = a^2 + 2ac$.

33-(MAT/10_HL_Winter_2020_Q8)-*Trigonometry*

(a) Show that $\dfrac{\sin x \tan x}{1-\cos x} \equiv 1+\dfrac{1}{\cos x}$, $x \neq 2n\pi$, $n \in \mathbb{Z}$. [3]

(b) Hence determine the range of values of k for which $\dfrac{\sin x \tan x}{1-\cos x} = k$ has no real solutions. [4]

34-(MathAA/12_HL_Summer_2021_Q2)**-***Trigonometry*

[Maximum mark: 7]

Solve the equation $2\cos^2 x + 5\sin x = 4$, $0 \le x \le 2\pi$.

35-(MathAA/11_HL_Summer_2021_Q5)**-***Trigonometry*

[Maximum mark: 8]

(a) Show that $\sin 2x + \cos 2x - 1 = 2\sin x(\cos x - \sin x)$. [2]

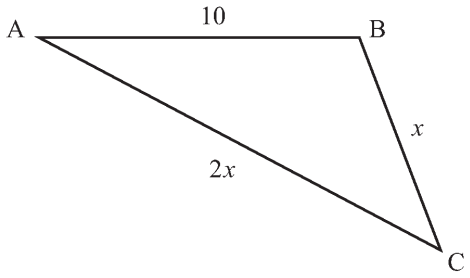
(b) Hence or otherwise, solve $\sin 2x + \cos 2x - 1 + \cos x - \sin x = 0$ for $0 < x < 2\pi$. [6]

36–(MathAA/12_HL_Summer_2021_Q6)–*Trigonometry*

[Maximum mark: 7]

The following diagram shows triangle ABC, with $\mathrm{AB} = 10$, $\mathrm{BC} = x$ and $\mathrm{AC} = 2x$.

diagram not to scale



Given that $\cos\hat{\mathrm{C}} = \frac{3}{4}$, find the area of the triangle.

Give your answer in the form $\frac{p\sqrt{q}}{2}$ where $p, q \in \mathbb{Z}^+$.

37-(MathAA/11_HL_Summer_2021_Q6)**-***Trigonometry*

[Maximum mark: 4]

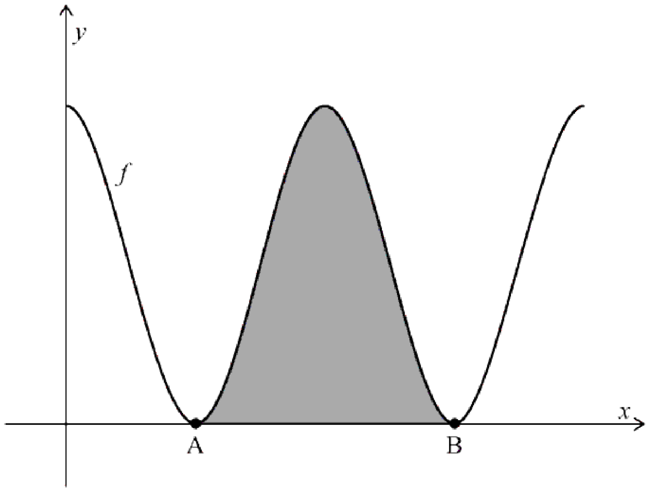
It is given that $\operatorname{cosec}\theta = \frac{3}{2}$, where $\frac{\pi}{2} < \theta < \frac{3\pi}{2}$. Find the exact value of $\cot\theta$.

38-(MathAA/12_HL_Summer_2021_Q10)-*Trigonometry*

[Maximum mark: 15]

Consider the function f defined by $f(x) = 6 + 6\cos x$, for $0 \leq x \leq 4\pi$.

The following diagram shows the graph of $y = f(x)$.



The graph of f touches the x-axis at points A and B, as shown. The shaded region is enclosed by the graph of $y = f(x)$ and the x-axis, between the points A and B.

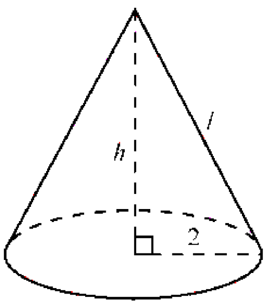
(a) Find the x-coordinates of A and B. [3]

(b) Show that the area of the shaded region is 12π. [5]

The right cone in the following diagram has a total surface area of 12π, equal to the shaded area in the previous diagram.

The cone has a base radius of 2, height h, and slant height l.

diagram not to scale



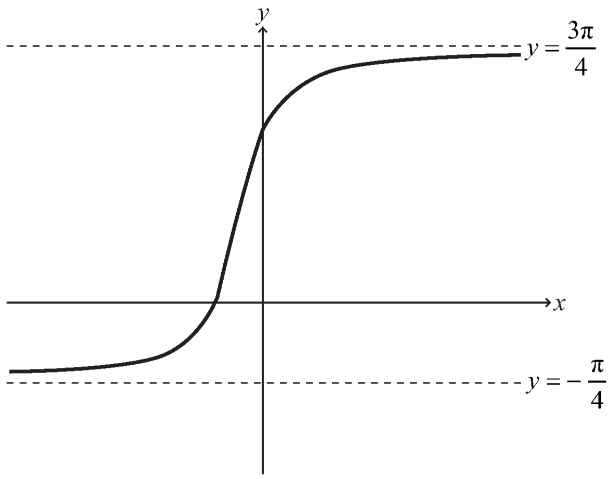
(c) Find the value of l. [3]

(d) Hence, find the volume of the cone. [4]

39-(MathAA/12_HL_Summer_2021_Q12)-*Trigonometry,MathematicalInduction*

[Maximum mark: 19]

The following diagram shows the graph of $y = \arctan(2x+1) + \frac{\pi}{4}$ for $x \in \mathbb{R}$, with asymptotes at $y = -\frac{\pi}{4}$ and $y = \frac{3\pi}{4}$.



(a) Describe a sequence of transformations that transforms the graph of $y = \arctan x$ to the graph of $y = \arctan(2x+1) + \frac{\pi}{4}$ for $x \in \mathbb{R}$. [3]

(b) Show that $\arctan p + \arctan q \equiv \arctan\left(\frac{p+q}{1-pq}\right)$ where $p, q > 0$ and $pq < 1$. [4]

(c) Verify that $\arctan(2x+1) = \arctan\left(\frac{x}{x+1}\right) + \frac{\pi}{4}$ for $x \in \mathbb{R}, x > 0$. [3]

(d) Using mathematical induction and the result from part (b), prove that $\sum_{r=1}^{n} \arctan\left(\frac{1}{2r^2}\right) = \arctan\left(\frac{n}{n+1}\right)$ for $n \in \mathbb{Z}^+$. [9]

40-(MathAA/10_HL_Winter_2021_Q6)-*Trigonometry*

[Maximum mark: 7]

(a) Show that $2x-3-\dfrac{6}{x-1}=\dfrac{2x^2-5x-3}{x-1}$, $x \in \mathbb{R}$, $x \neq 1$. [2]

(b) Hence or otherwise, solve the equation $2\sin 2\theta-3-\dfrac{6}{\sin 2\theta-1}=0$ for $0 \leq \theta \leq \pi$, $\theta \neq \dfrac{\pi}{4}$. [5]

41-(MathAA/12_HL_Summer_2022_Q4)-*Trigonometry*

[Maximum mark: 5]

Find the least positive value of x for which $\cos\left(\frac{x}{2}+\frac{\pi}{3}\right)=\frac{1}{\sqrt{2}}$.